9.6.5 Wrap-Up: Growth Mindset

"You may encounter many defeats, but you must not be defeated. In fact, it may be necessary to encounter the defeats, so you can know who you are, what you can rise from, how you can still come out of it." — Maya Angelou

1. What was the toughest part of today's lesson for you?

2. How did you keep going when things got tough?

3. Write a short note to yourself or draw a picture that celebrates your perseverance through the tough part of today's lesson.

Unit 9, Lesson 7: Solving Quadratics by Factoring

Benchmark

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

Learning Target

- [ ] I can solve a quadratic by factoring.

9.7.1 Warm-Up: Satellite Dishes

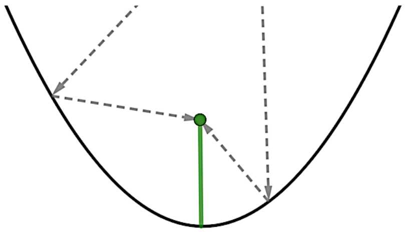
Valerie Thomas is a scientist and inventor. In 1980, she received a patent for inventing the illusion transmitter. This was an integral piece of technology in developing the digital media formats needed for today's video games, movies, and 3D television.

Thomas worked at NASA from 1964-1995, where she helped develop real-time computer data systems to support satellite operations. She oversaw the creation of the Landsat program there, which supported the first satellite to transmit images from outer space.

Today, satellite dishes are everywhere. Satellite dishes are designed using a parabolic curve.

1. What key feature of a parabola is the bottom of the dish?

2. The arrows show how signals bounce off a satellite dish to the receiver. The receiver is always located at a parabola's focus. What key feature does the focus lie on?



9.7.2 Activity: Card Sort

1. Work with your partner to match the cards.

Equation	Graph	x-intercept	x-intercept
P			
Q			
R			
S			

2. Factor each expression.

Expression T	Expression U	Expression V	Expression W
$2x^2 + 5x - 42$	$2x^2 - 5x - 42$	$2x^2 - 19x + 42$	$2x^2 + 19x + 42$

3. Work with your partner to match the factored expressions to the x-intercepts.

Expression	Factored Expression	x-intercepts
T		
U		
V		
W		

4. Discuss with your partner how the factored expression reveals the x-intercepts. Make a conjecture about the connection.

9.7.3 Guided Instruction: Solving Quadratic Equations by Factoring

1. In the equation $x \cdot y \cdot z = 0$, which variable has to equal 0 so that the product is equal to 0?

- ○ x
- ○ y
- ○ z
- ○ At least one of x, y, or z

Zero Product Property

If the product of two numbers is 0, then at least one of the numbers is 0.

2. For each expression, use the zero product property to determine the values of x for which the expression would equal 0.

Expression	$(x+7)(x-2)$	$(2x+7)(x-6)$	$(-3x+11)(4x+18)$
x-values			

Each of these expressions could have been written as quadratic equations that are set equal to 0 such as $x^2+5x-14=0$.

3. When solving a quadratic equation set the equation equal to 0, the solution(s) will be the value(s) of x when $y =$______. Explain how this relates to your conjecture from the card sort.

Another method of solving quadratics is by factoring. When solving by factoring the quadratic must be set equal to 0. Once the quadratic is factored, the zero product property is applied.

4. Solve $h^2 - 10h + 25 = 0$ by factoring.

5. Solve $10f^2 + 13f - 3 = 0$ by factoring.

To solve by factoring, the quadratic equation should be in standard form and set equal to zero.

6. Solve $p^2 + 36 = 13p$ for p by factoring.

7. Solve $40x^2 - 18x = 12x + 135$ for x by factoring.

9.7.4 Try It: Factoring

Solve each by factoring. Show your work.

1. $x^2 - 10x - 15 = 9$

2. $-3x^2 + 6x + 22 = 2 - 5x$

9.7.5 Wrap-Up: Error Analysis

The work of two students who solved $2x^2 - 12x + 14 = 2x - 6$ is shown.

Ruchelle	Quinyang
$2x^2 - 12x + 14 = 2x - 6$	$2x^2 - 12x + 14 = 2x - 6$
$2(x^2 - 6x + 7) = 2(x - 3)$	$2x^2 - 12x + 14 - (2x - 6) = 2x - 6 - (2x - 6)$
$\frac{2(x^2 - 6x + 7)}{2} = \frac{2(x - 3)}{2}$	$2x^2 - 14x + 20 = 0$
$x^2 - 6x + 7 = x - 3$	$(2x - 5)(x - 4) = 0$
$x^2 - 5x + 4 = 0$	$x = \frac{5}{2}$ or $x = 4$
$(x - 4)(x - 1) = 0$	
$x = 4$ or $x = 1$	

1. Both Ruchelle and Quinyang made one mistake. Circle each mistake.